Launch Vehicle Simulator

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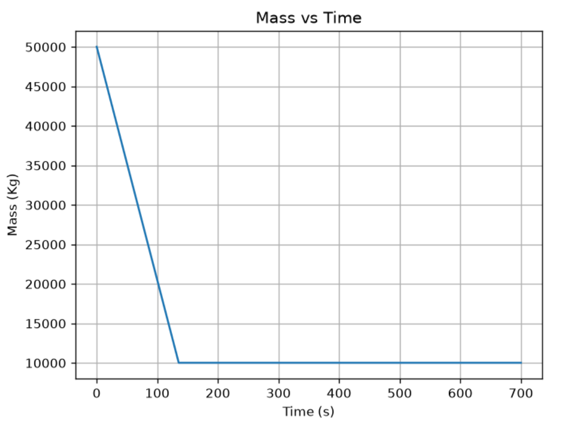
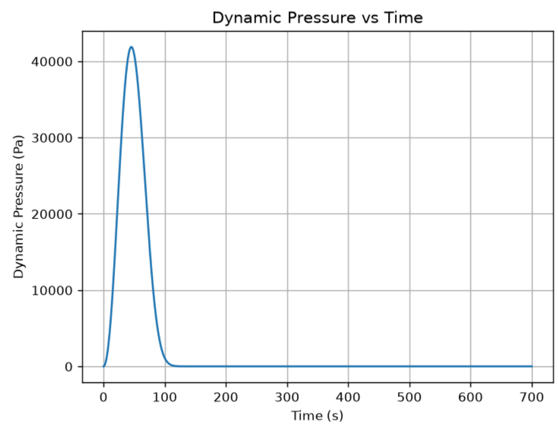
The goal of this project is to utilize python to create a simulation of a simple rocket launching from earth and plotting the various outputs in accordance with time. The focus is on ascent part of the launch, with calculations also working for the early part of the decent process. The graphs showcase the maximum dynamic pressure on the vehicle and its performance metrics as it climbs through the atmosphere.

The calculations were started off by creating the file layout in python starting with given data, values were estimated for this exercise but can easily be applied to any launch vehicle. This includes wet and dray mass, burn time, specific impulse, and drag coefficient. Other basics like acceleration of gravity and atmosphere density were included. The density was set to the sea level value for the first calculation. Since there is less density as the rocket travels upward a program created to account for the change in the calculations, making them more accurate.

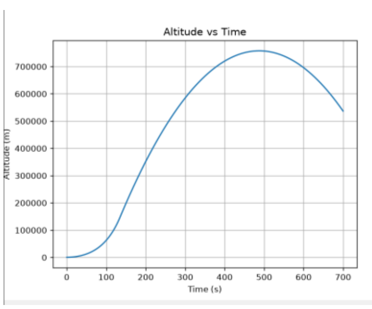
The mass of the rocket is constantly changing throughout the whole process and has a major impact on the calculations. For simplistic purposes the rocket used has a constant thrust throughout the whole burn phase, so it will lose its fuel mass at a constant rate. The code created under vehicle reflects this by updating the vehicle mass for the given time interval of a tenth of a second. This plays a critical role in the force calculations program with uses lift, drag and mass. The lift is constant but the drag and mass decrease from the fuel burning and the thinner atmosphere respectively. This change in forces is needed to show the increase in acceleration as the rocket goes higher, reflecting on known rocket principles.

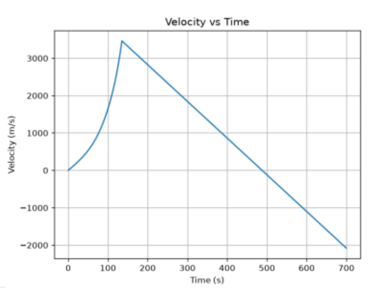
All the equations used are then fed into the simulation that calculates mass, velocity, acceleration, altitude and dynamic pressure. This is done at an interval of a tenth of a second to avoid gaps in plotting and ensure that there is enough information. This loop runs for a period of 700 seconds or until the altitude is less then zero meters. All the values collected through this process are then plotted compared to time and produce the graphs seen below.

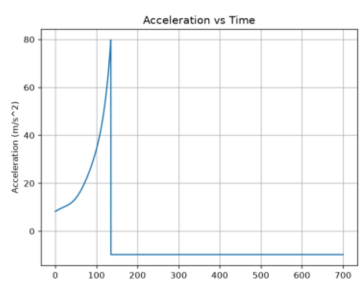




Since the thrust is constant the mass decreases constantly as seen in graph. The main point of focus is the dynamic pressure which shows the maximum aerodynamic load on the vehicle. Based on the graph, the highest point is around 45 seconds into the launch. The total burn time for the launch is 135 seconds. This happens when the rocket is still at a low point in the atmosphere and happens at a time compared to rockets launching today. The main difference is that it is high due to the steady thrust, normally less thrust is produced here so that the rocket encounters less load.







The altitude, velocity and acceleration graphs are being looked at together because they are essentially derivatives of each other. The rocket clearly starts off slowly for many reasons, it has its highest mass and the most atmospheric drag in its early stages. This is shown by a slower velocity and in the early first few seconds. The main point of change is after the maximum dynamic pressure point talked about above. The altitude increases at a much faster rate along with velocity and acceleration. There is a significant increase in the jerk of the rocket after the dynamic pressure point because of the less forces slowing it down.

Momentum plays a major role here too, with the rocket continuing to increase in altitude for around 500 seconds. The rocket continues to move upwards for this period as it starts to accelerate back down to earth. The accelerations graph the shows the rapid drop that takes place, but the velocity is still positive until the 500 second mark. The rocket is slowing down even though it is still traveling upward and will continue to do so long last the burnout time. This is because there are little to no drag forces acting on the rocket when it is in space, the acceleration due to gravity is pulling it down.

While the graphs did produce accurate representations of the launch, there are many constant factors with will skew the data. The main one being vertical flight all the way into orbit, which produced a higher stress on the rocket. Normally there would be a pitch following liftoff so that the rocket can fly horizontal and have an easier time slicing through the atmosphere. These calculations are also for a simple atmosphere which doesn't consider all the subtle change that are present in different regions of the Earth. This creates a big difference for precise calculations and effects the data output. Future variations can take these constant factors and adjust them to make the model more accurate.

Overall, the project was successful in modeling the altitude, velocity, acceleration, and dynamic pressure of a simple rocket over a period of 700 seconds. Producing accurate and relevant data for the values given during the time frame. Python code was successful in calculating and modeling the principles of rocket flight in an organized manor.